Name _____

PRACTICE
CHM 2410
Exam 3 Fall 2025

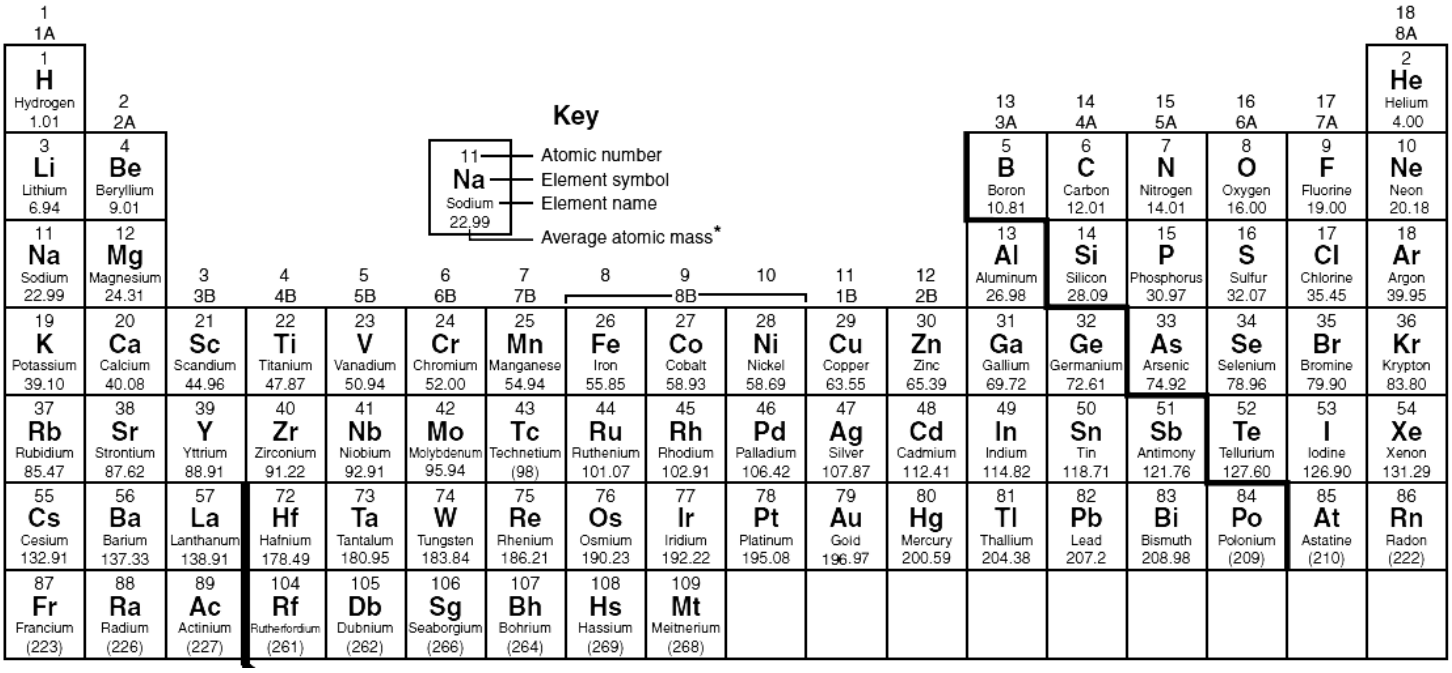
Notes:
1. The test is double-sided.
2. Carefully read the directions above each section of the exam.
3. Mark all answers clearly and **show all work** for possible partial credit.

**Good luck!**

I pledge that I have neither given nor received aid on this test:

_____

(Signature)




| | 1 1A | | | | | | | | | | | | 13 3A | 14 4A | 15 5A | 16 6A | 17 7A | 18 8A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 H Hydrogen 1.01 | 2 2A | | | | | | | | | | | | | | | | 2 He Helium 4.00 |
| 2 | 3 Li Lithium 6.94 | 4 Be Beryllium 9.01 | | | | | | | | | | | 5 B Boron 10.81 | 6 C Carbon 12.01 | 7 N Nitrogen 14.01 | 8 O Oxygen 16.00 | 9 F Fluorine 19.00 | 10 Ne Neon 20.18 |
| 3 | 11 Na Sodium 22.99 | 12 Mg Magnesium 24.31 | 3 3B | 4 4B | 5 5B | 6 6B | 7 7B | 8 | 9 8B | 10 | 11 1B | 12 2B | 13 Al Aluminum 26.98 | 14 Si Silicon 28.09 | 15 P Phosphorus 30.97 | 16 S Sulfur 32.07 | 17 Cl Chlorine 35.45 | 18 Ar Argon 39.95 |
| 4 | 19 K Potassium 39.10 | 20 Ca Calcium 40.08 | 21 Sc Scandium 44.96 | 22 Ti Titanium 47.87 | 23 V Vanadium 50.94 | 24 Cr Chromium 52.00 | 25 Mn Manganese 54.94 | 26 Fe Iron 55.85 | 27 Co Cobalt 58.93 | 28 Ni Nickel 58.69 | 29 Cu Copper 63.55 | 30 Zn Zinc 65.39 | 31 Ga Gallium 69.72 | 32 Ge Germanium 72.61 | 33 As Arsenic 74.92 | 34 Se Selenium 78.96 | 35 Br Bromine 79.90 | 36 Kr Krypton 83.80 |
| 5 | 37 Rb Rubidium 85.47 | 38 Sr Strontium 87.62 | 39 Y Yttrium 88.91 | 40 Zr Zirconium 91.22 | 41 Nb Niobium 92.91 | 42 Mo Molybdenum 95.94 | 43 Tc Technetium (98) | 44 Ru Ruthenium 101.07 | 45 Rh Rhodium 102.91 | 46 Pd Palladium 106.42 | 47 Ag Silver 107.87 | 48 Cd Cadmium 112.41 | 49 In Indium 114.82 | 50 Sn Tin 118.71 | 51 Sb Antimony 121.76 | 52 Te Tellurium 127.60 | 53 I Iodine 126.90 | 54 Xe Xenon 131.29 |
| 6 | 55 Cs Cesium 132.91 | 56 Ba Barium 137.33 | 57 La Lanthanum 138.91 | 72 Hf Hafnium 178.49 | 73 Ta Tantalum 180.95 | 74 W Tungsten 183.84 | 75 Re Rhenium 186.21 | 76 Os Osmium 190.23 | 77 Ir Iridium 192.22 | 78 Pt Platinum 195.08 | 79 Au Gold 196.97 | 80 Hg Mercury 200.59 | 81 Tl Thallium 204.38 | 82 Pb Lead 207.2 | 83 Bi Bismuth 208.98 | 84 Po Polonium (209) | 85 At Astatine (210) | 86 Rn Radon (222) |
| 7 | 87 Fr Francium (223) | 88 Ra Radium (226) | 89 Ac Actinium (227) | 104 Rf Rutherfordium (261) | 105 Db Dubnium (262) | 106 Sg Seaborgium (266) | 107 Bh Bohrium (264) | 108 Hs Hassium (269) | 109 Mt Meitnerium (268) | | | | | | | | | |

Name _____

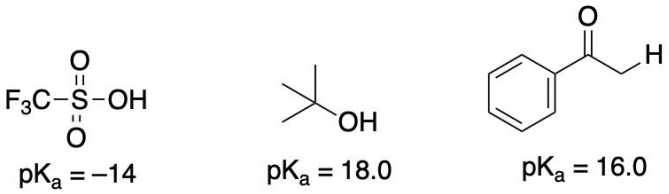
**Multiple Choice Section:** Choose the *best* answer and write the corresponding letter in the space provided (3 pts each, 45 points total). ____/45

_____**9.** Which of the following best explains which compound is more acidic?

HF or HBr

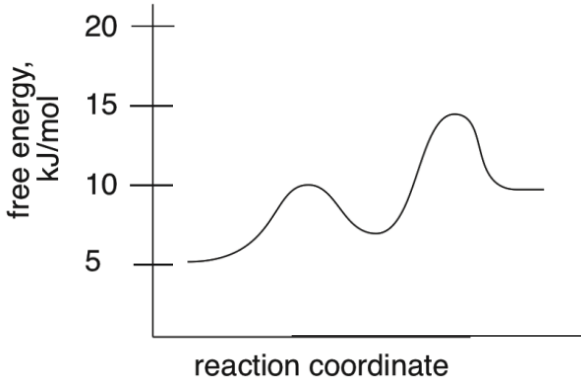
a. HF is more acidic due to the F atom being more electronegative than Br
b. HBr is more acidic due to the Br atom being less electronegative than F
c. HF is more acidic due to the $F^-$ anion being smaller than $Br^-$
d. HBr is more acidic due to the $Br^-$ anion being larger than $F^-$
e. HF is capable of participating in hydrogen bonding and therefore is a strong acid

_____**13.** Which of the following is the strongest acid?




**A** **B** **C**

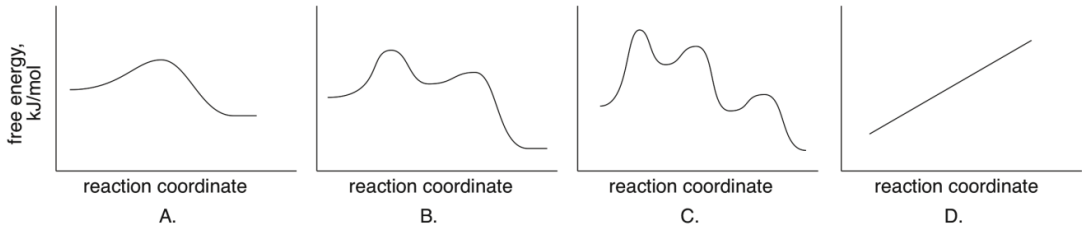
____ 3. Which is true about the following energy diagram?



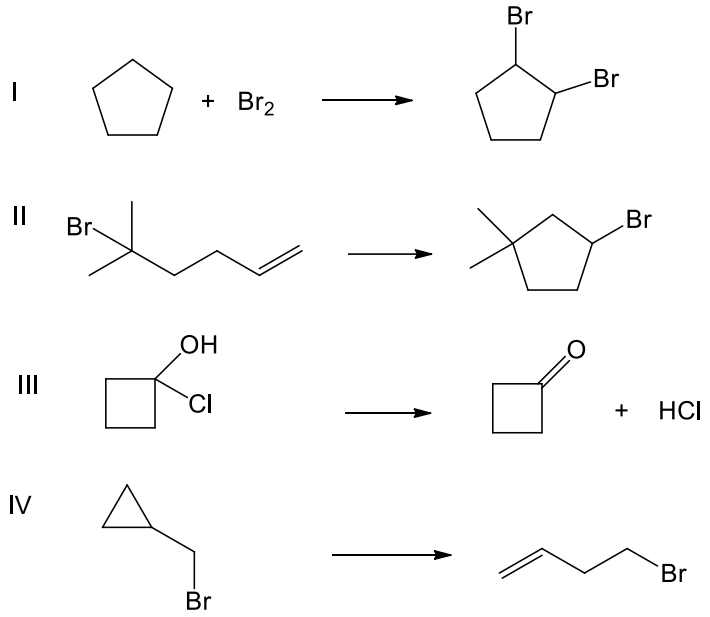

a) The energy diagram is depicting a reaction that is nonspontaneous because the free energy of the products is greater than the free energy of the reactants
b) The energy diagram is depicting a reaction that is spontaneous the free energy of the products is greater than the free energy of the reactants
c) The energy diagram is depicting a reaction that is nonspontaneous because there are two transition states.
d) The energy diagram is depicting a reaction that is spontaneous because it only has a 5kJ activation energy.

____ 7. Which of the following is an energy diagram with 1 transition state?

Name _____



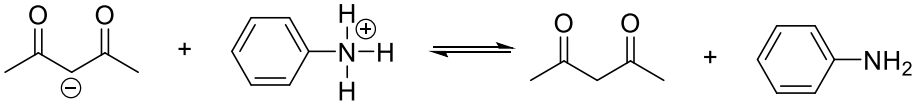

____ 13. From the following reactions, which one(s) would you expect to have a negative ΔS?




a. I
b II
c. III
d. IV
e. I and II
f. III and IV

**Part II: Short Answer Section** Explain your answers. _____/62

**17. a)**Identify the acid, base, conjugate acid, and conjugate base in the following reaction. (4 pt)




21. Answer the following questions about ice melting.

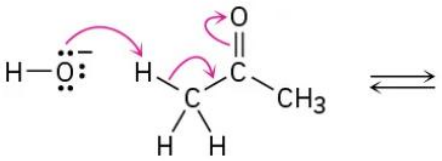
a) Is ice melting spontaneous at room temperature? _____ (2 pt)

b) When ice melts does entropy increase or decrease? _____ (2 pts)

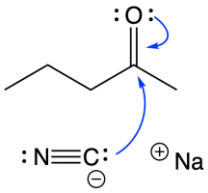
c) Is ice melting spontaneous at all temperatures? Explain your answer (4 pts)

22. Draw the product of the following arrows. (5 pt)

Name _____



23. The following has incorrectly drawn arrows. Explain why they are wrong. (3 pt)

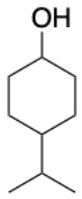


24. a) Does breaking bonds release energy or take energy? _____ (3 pts)

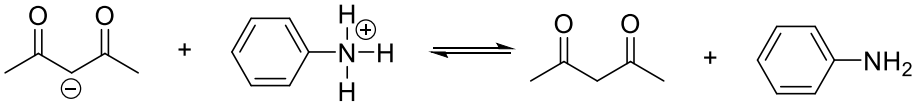
b) Based on the answer to (a) is breaking bonds a exothermic or endothermic process?

_____ (3 pts)

**16.** Draw the conjugate acid for the following base. (4 pts)

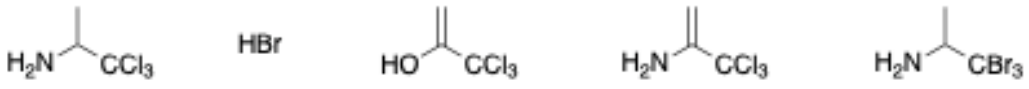


**23)** In the reaction below draw in ALL the arrows on the **REACTANT side** indicating how the reaction happed. (4 pts)



b) Compare the acid and the conjugate acid and determine which side the equilibrium would lie (reactants or products). Explain your answer.

24. a) Rank the following acids (2 pts)



**A** **B** **C** **D** **E**

#1(Most Acidic) _____; #2_____; #3_____; #4_____; #5_____(Least Acidic)

b) Explain ALL your reasoning (6 pts)